jigsaw-agile-community-rules

5th Place solution - Diverse Ensemble

Rank	5
Team	Flash ⚡
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First of all thanks to the organizers, wonderful public notebooks, and the community and everyone who shared ideas along the way.

Overview

Our final solution is a deliberately diverse ensemble. We combined per-rule specialists (DeBERTa NLI training; embedding models with logistic regression + k-NN), Inference-only large model (Qwen-32B trained on the official data and inferred with similarity-based sampling), and several test-time-trained (TTT) components across all rules (public DeBERTa training; public FAISS + triplet pipeline; Qwen-3 1.7B full finetune; Llama-3.1 8B LoRA). All models used a consistent instruction-only prompt and Yes/No-constrained decoding, then we blended them with per-rule rank-normalized weighting.

Data Preprocessing

- Drop `subreddit` - Treated as noisy/unstable signal; removing it improved consistency.
- Deduplicate rows: Removed exact duplicates, reducing train size to <10k examples, faster training with no measurable drop on lb.

Performance Evaluation

We used the “trust public LB” method.

Models

Model	Public LB	Private LB	Runtime*
LLaMA-3.1 8B Instruct (TTT)	0.92084	0.92618	~130 min
Qwen-3 8B (TTT)	0.91971	0.92314	~130 min
Triplet + FAISS (public, improved)	0.90082	0.91131	~70 min
6× Embedding models → LR + kNN (per-rule)	0.90067	0.90976	~90 min
DeBERTa (public training recipe)	0.90149	0.90767	~45 min
DeBERTa NLI (custom)	0.89853	0.90636	~90 min
Qwen-32B (Inference-only, similarity sampling)	0.89065	0.89671	~300 min

* End-to-end runtime on our 2×T4 setup.

Training and Inference:

LLaMA-3.1 8B Instruct & Qwen-3 8B (LoRA, DDP)

- Setup: 2×T4 with DDP, fp16 + grad checkpointing, seq len 512 for train + inference.
- LoRA: `r=256, lora_alpha=32, lora_dropout=0.0, bias="none", target_modules=["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"].`
- Opt/Schedule: paged_adamw_8bit, LR≈2e-4, cosine, warmup≈10%.

Prompt :

```
Instructions: You are a expert Reddit Moderator, you are given a comment and a rule. Your task is to classify whether the comment violates the rule. Only respond Yes/No. <Rule>{rule}</Rule> <Comment>{body}</Comment> Answer:
```

Inference:

vLLM TP=2, max_model_len=512; constrained to {Yes, No}, max_tokens=1, take Yes prob; if over budget, truncate only .

Embedding Models

Families used: E5 (base, large), Qwen3-Embedding (0.6B), BGE (small, large), GTE (large).

Per-rule pipeline

- De-dup & de-bias: cluster near-duplicates within each rule (cosine > 0.95–0.97) and down-weight them so the classifier doesn't overfit repeated patterns.
- Train two light heads (6-fold CV):
  - Logistic Regression: grid over 31 C values .
  - KNN: grid over K=10–15, distance-weighted.
  - For each embed model & rule, keep the best CV setting.
- Head blending (per rule): 0.7 · LR + 0.3 · KNN to couple calibration (LR) with local neighborhood cues (KNN).
- Model ensemble search (per rule): GPU-accelerated simplex/grid over weights for the six embed models to maximize CV AUC (constrained, non-negative, sum to 1).
- Stabilization: final per-rule score = 0.9 · ensemble probability + 0.1 · rank-mean (guards against tail bias / scale drift across rules).
- Output: calibrated probability per row, per rule—later blended with other model families in the overall ensemble.
  (This embed stack was our earliest strong baseline; we iterated heavily on de-dup thresholds, CV folds and embed models)

DeBERTa NLI (rule-aware zero-shot → TTT)

Base: `MoritzLaurer/deberta-v3-large-zeroshot-v2.0`
NLI framing:
- Premise: comment body
- Hypothesis: “This comment violates the rule: {rule}.” (plus 8 paraphrases)
- Label map: Entailment → violation, Contradiction/Neutral → compliant

Per-rule Test-Time Training (TTT):
- Build a small rule-specific set from test canonical positives/negatives.
- Upsample to ~25k pairs; train 2 epochs.
- Update only biases + classifier head + LayerNorms (backbone frozen), faster training.
Balancing: dynamic pos/neg multipliers to ~50/50 (cap 32×).

Qwen 2.5 32B LLM with Asymmetric Example Selection

Core idea: Before each inference, pick contextually relevant examples using embeddings—then prompt the LLM.

- Positive examples: choose the most similar to the comment (clear violation prototypes).
- Negative examples: choose the least similar (hard, diverse non-violations).
  This asymmetric pairing tightens the decision boundary.

Pipeline:

1. Embed canonicals (2 pos + 2 neg per rule) with Qwen3-0.6B embeddings.
2. For each comment: encode (rule+comment) → compute cosine sims → argmax pos, argmin neg, with simple quality thresholds.
3. Build the prompt with the selected pos/neg examples.

Infer with Qwen-32B-Instruct (GPTQ) + LoRA; constrain outputs to Yes/No and softmax logits.
Result: +0.006 vs vanilla few-shot—showing that relevant positives + hard negatives noticeably improve classification.

Triplet + FAISS

Include the train set in public solution.
Train for extra epoch

Ensemble (stack-as-you-go, per-rule ranks)

We ensembled incrementally, whenever a new strong model landed, we stacked it on top of the current blend rather than rebuilding from scratch. Scores from each submission were rank-normalized per rule to remove calibration drift across models (LLMs vs. NLI vs. retrieval). We then applied a hierarchical blending schedule: start with a 50/50 of the strongest retrieval (Embeddings LR+kNN) and the strongest NLI (DeBERTa-NLI), and progressively mix in FAISS/Triplet, Qwen-32B (train-only), DeBERTa-public, Qwen-3-8B (TTT), and LLaMA-3.1-8B (TTT). Each addition used a fixed mixing ratio; the previous stack was down-weighted accordingly (geometric decay), which naturally regularizes and prevents any single model from dominating. Finally, we clipped to [1e-6, 1−1e-6] . This strategy leveraged diversity across paradigms (retrieval, NLI, LLM-TTT) and kept calibration stable.

Citation

https://www.kaggle.com/code/neibyr/lb-0-878-train-on-test-data-qwen2-5-0-5b
https://www.kaggle.com/code/nahidhossainredom/deberta-v3-base-3-epochs-lb-0-906
https://www.kaggle.com/code/datafan07/jigsaw-speed-run-10-min-triplet-and-faiss